

Model: LightGBM & XGBoost(Team B)

1. Objective

To build and evaluate machine learning models to predict the `actual\_time` based on engineered features.

Two powerful gradient boosting algorithms are compared:

- LightGBM (Light Gradient Boosting Machine)
- XGBoost (Extreme Gradient Boosting)

2. Data Preparation

- Dataset: feature_engineered_dataset.csv
- Removed rows with missing `actual\_time`.
- Imputed missing values:
 - Mean for numeric features (LightGBM pipeline)
 - Median + OneHot for categorical features (XGBoost pipeline)
- Replaced infinite values with NaN.
- Applied log1p transformation on target in XGBoost to reduce skew.
- Train/Test Split: 80% training, 20% testing.

3. Model Training & Setup

LightGBM:

- Simple pipeline using numeric features only.
- No scaling or encoding required.

XGBoost:

- Full preprocessing pipeline with scaling and one-hot encoding.
- GridSearchCV for hyperparameter tuning (n_estimators, learning_rate, etc.).
- Applied log transformation to target and inverse transform for final prediction.

4. Performance Evaluation

Models were evaluated using the following metrics:

- RMSE (Root Mean Squared Error)

- MAE (Mean Absolute Error)
- R² Score (Coefficient of Determination)

Model Evaluation Metrics

Metric	XGBoost Train	XGBoost Test	LightGBM Train	LightGBM Test
RMSE	39.18	34.98	7.18	8.06
MAE	11.92	11.37	4.17	4.37
R² Score	0.9960	0.9966	0.9999	0.9998

5. Interpretation & Insights

- LightGBM outperformed XGBoost in all key metrics (lower error, higher R²).
- LightGBM required less preprocessing and training time.
- XGBoost performance was strong but slightly weaker, possibly due to target transformation.
- Feature importance visualizations helped identify critical predictors (done in LightGBM pipeline).

6. Conclusion & Recommendation

LightGBM is the best choice for this regression task.

- Delivers lower prediction error and higher accuracy.
- Simplified pipeline and less complexity.
- Ideal for fast deployment and scalability.